

MADISON, SD, USA

WMO: 720624

Lat: 44.016N

Lon: 97.086W

Elev: 1716

StdP: 13.81

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00211

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.0	-17.2	2.3	-10.6	-11.4	3.2	-6.9	34.6	19.6	31.8	17.4	10.2	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	88.3	74.4	84.4	72.8	82.1	71.3	79.2	83.5	76.8	80.9	74.8	79.3	14.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	152.4	80.2	75.3	141.2	78.3	73.1	131.2	76.8	44.1	82.9	41.8	81.1	39.8	79.5	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.0	26.1	23.6	-21.6	92.2	5.3	2.4	-25.4	93.9	-28.5	95.3	-31.5	96.7	-35.4	98.4		
			-20.6	83.9	6.3	2.4	-25.2	85.6	-28.9	87.0	-32.4	88.4	-37.0	90.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	15.2	15.7	31.9	43.5	56.4	67.0	70.6	67.5	61.4	46.3	32.7	19.1
	DBStd	22.29	13.11	12.21	13.40	10.42	9.38	6.54	5.95	5.84	7.72	9.29	11.06	11.81
	HDD50	4572	1077	961	576	242	46	0	0	0	8	183	522	957
	HDD65	8132	1542	1381	1026	645	297	54	17	41	157	582	968	1422
	CDD50	2417	0	0	15	48	243	509	637	543	350	68	4	0
	CDD65	502	0	0	1	1	29	112	190	118	49	2	0	0
	CDH74	4325	0	0	8	39	338	985	1673	884	368	30	0	0
	CDH80	1212	0	0	1	10	97	285	512	228	74	5	0	0
Wind	WSAvg	11.4	12.5	12.3	12.5	13.6	12.5	10.7	8.9	8.4	10.1	11.5	12.1	12.2
Precipitation	PrecAvg	26.7	0.6	0.7	1.2	2.5	3.5	4.6	3.5	3.1	3.0	2.2	1.0	0.7
	PrecMax	35.6	1.3	1.5	3.5	6.0	5.9	9.5	6.8	5.9	7.3	5.8	3.1	1.9
	PrecMin	18.6	0.0	0.2	0.2	0.6	1.3	1.8	0.6	0.7	0.4	0.2	0.0	0.0
	PrecStd	4.2	0.4	0.4	0.8	1.3	1.5	2.2	1.8	1.5	1.8	1.6	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	55.1	73.1	80.6	88.5	90.8	91.4	90.1	86.0	78.9	64.4	49.8
		MCWB	39.2	48.8	56.3	60.1	65.5	73.3	77.5	76.4	71.6	62.2	52.8	45.0
	2%	DB	38.5	44.7	63.9	72.0	81.9	86.1	88.5	84.5	81.9	72.1	57.2	42.7
		MCWB	36.3	40.5	53.8	55.3	64.2	71.5	76.4	74.8	70.0	58.4	47.2	37.8
	5%	DB	35.9	37.1	57.1	65.6	77.0	82.4	85.5	82.0	78.6	66.1	52.5	37.0
		MCWB	34.9	35.7	50.3	51.9	61.9	69.5	75.5	73.4	67.6	55.4	45.6	34.7
	10%	DB	32.3	32.4	50.2	60.6	72.7	80.7	82.1	79.1	74.8	62.5	47.7	34.1
		MCWB	32.0	31.7	45.8	49.6	60.3	68.8	73.2	71.0	67.3	53.6	42.9	32.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	50.1	62.8	62.8	72.7	81.0	84.0	81.2	76.9	64.1	53.9	45.8
		MCDB	44.9	53.3	66.8	69.7	76.8	83.7	86.4	82.5	80.0	70.0	62.3	47.5
	2%	WB	36.8	40.0	55.2	58.7	68.1	76.5	81.1	77.4	74.1	60.7	49.7	38.3
		MCDB	38.2	42.4	60.6	67.1	75.7	81.1	83.3	81.2	77.7	68.2	56.4	41.3
	5%	WB	34.3	36.2	50.3	55.3	65.8	73.5	77.4	75.5	72.1	57.5	46.1	35.8
		MCDB	35.4	38.1	57.8	61.4	74.5	78.6	81.5	79.8	75.7	64.1	50.2	37.8
	10%	WB	32.2	32.2	45.6	52.1	63.3	71.2	75.3	73.4	68.4	54.3	43.0	32.4
		MCDB	32.8	33.1	50.1	59.7	68.1	76.8	80.1	76.7	73.1	60.1	48.0	33.8
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.8	17.4	19.4	20.0	18.9	19.5	19.1	20.2	19.1	17.8	14.5
		MCDBR	18.6	20.4	28.4	30.1	26.2	22.6	21.2	21.3	23.6	26.9	24.6	19.0
		MCWBR	16.4	16.2	18.1	16.1	13.1	11.9	11.5	12.4	12.1	15.0	15.0	15.2
	5% WB	MCDBR	17.9	20.1	27.4	23.4	21.4	19.3	18.8	18.1	19.5	23.1	21.7	18.4
		MCWBR	16.2	16.4	18.6	15.8	12.9	13.4	12.7	12.7	13.4	14.6	14.5	15.3
Clear-Sky Solar Irradiance	taub	0.269	0.284	0.313	0.364	0.378	0.388	0.393	0.397	0.350	0.320	0.287	0.267	
	taud	2.423	2.404	2.413	2.295	2.339	2.367	2.367	2.347	2.317	2.429	2.486	2.486	2.442
	Ebn at Noon	276	290	294	285	283	280	277	272	278	273	267	265	
	Edh at Noon	23	28	32	39	39	38	38	38	31	26	22	21	
All-Sky Solar Radiation	RadAvg	597	930	1214	1541	1788	1998	2129	1758	1392	931	634	468	
	RadStd	58	77	109	141	131	137	76	137	104	110	53	39	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page